

Kathleen Alves Costa

[Email](#) | [Website](#) | [LinkedIn](#)

Skills

Programming & Scripting: Python, Perl, Prolog, Bash, Git, Linux command line

Machine Learning & NLP: spaCy, NLTK, scikit-learn, NumPy, pandas

Speech Technology: VITS (currently learning), Tacotron 2, FastSpeech, DeepSpeech, GlowTTS

Experience

Generative AI Chatbot Developer, Community School of the Arts Foundation – Remote July 2025 – Present

- Supported AI curriculum and generative content initiatives for K–12 arts education, integrating LLM-based tools and Canvas LMS for enhanced learning experiences.
- Developed and documented AI-driven educational resources using NLP, prompt engineering, AWS, React, HTML, and GitHub.
- Collaborated remotely via Slack and Zoom, maintaining project deadlines, version control, and adherence to data privacy and inclusive teaching practices.

Language Specialist/Quality Assurance Specialist, Innodata, Inc. – Remote May 2024 – Present

- Provided linguistic expertise and QA for AI training data in Portuguese and English
- Audited multilingual training data for quality, cultural relevance, and guideline compliance
- Managed multiple concurrent annotation projects with consistent on-time delivery

ESL Instructor, The Learning Center – Marietta, GA Sep 2023 – Dec 2023

- Developed and taught beginner-level English lessons for Spanish- and Portuguese-speaking adults
- Emphasized pronunciation, intonation, and oral comprehension
- Applied communicative and task-based approaches aligned with L2 speech acquisition research

Activities Intern, Georgia State University - IEP Program – Atlanta, GA May 2023 – July 2023

- Organized cultural and conversational ESL activities for international students
- Promoted events using multilingual flyers and social media (Canva, Instagram)
- Facilitated communication between students and instructors to support learning outcomes

Projects

Brazilian Portuguese TTS with Tacotron2 <https://git.new/portuguesetts>

- Developed a TTS pipeline using Tacotron 2 and FastSpeech trained on ~11 hours of data
- Used HPC cluster to train models across 250–2,500 epochs
- Verified audio quality through spectrogram analysis in Praat

Esperanto ASR with DeepSpeech <https://git.new/esperantostt>

- Trained ASR model for Esperanto using DeepSpeech on an HPC system
- Compared greedy vs. non-greedy decoding; non-greedy produced higher accuracy
- Final model generated intelligible output after 1,000 epochs

Webcam Dictation Assistant (*in progress – Summer 2025*)

- Real-time OCR-to-speech system using OpenCV, Tesseract, and VITS
- Captures webcam text and converts it into natural-sounding speech
- Adds accessibility features via langdetect and deep-translator
- Designed for users with visual impairments, dyslexia, or second-language needs

Education

University of Arizona – M.S. in Human Language Technology

Georgia State University – B.A. in Applied Linguistics